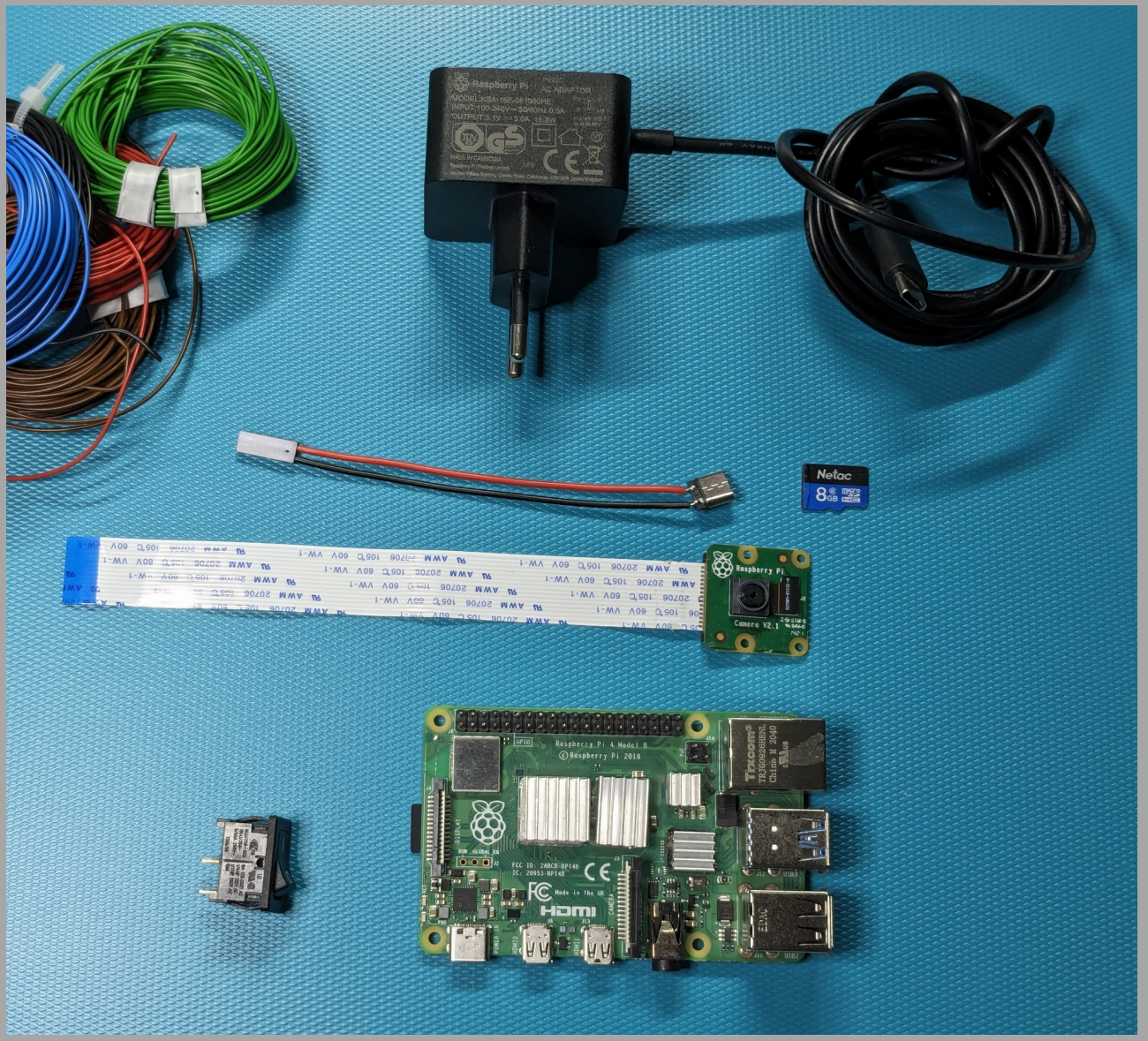


TERMINATOR SKULL

DESIGN BY LAYERLUX3D

This instruction is for people who build the T-800 skull without the face tracking and want to add it afterwards.

In the instruction the Raspberry Pi will be refered as RPI.



Those are the parts I used for the upgrade.

Note that if you use different parts the face tracking may not work as intended.

USB-C port

5V USB-C power supply with 3A

Raspberry Pi 4 Model B

Raspberry Pi Camera V2.1

8GB Micro SD card

Switch

Description how the Arduino and Raspberry Pi (RPI) interact.

The Arduino boots much faster than the RPI. Because of this, the Arduino controls the LEDs and servos. While the RPI starts, the Arduino plays the startup sequence where the eyes flicker and the head raises slowly.

After that, the Arduino regularly checks the GPIO pins of the RPI.

During the RPI's startup, the GPIO pins may already be active. These active GPIO pins are not valid face tracking values.

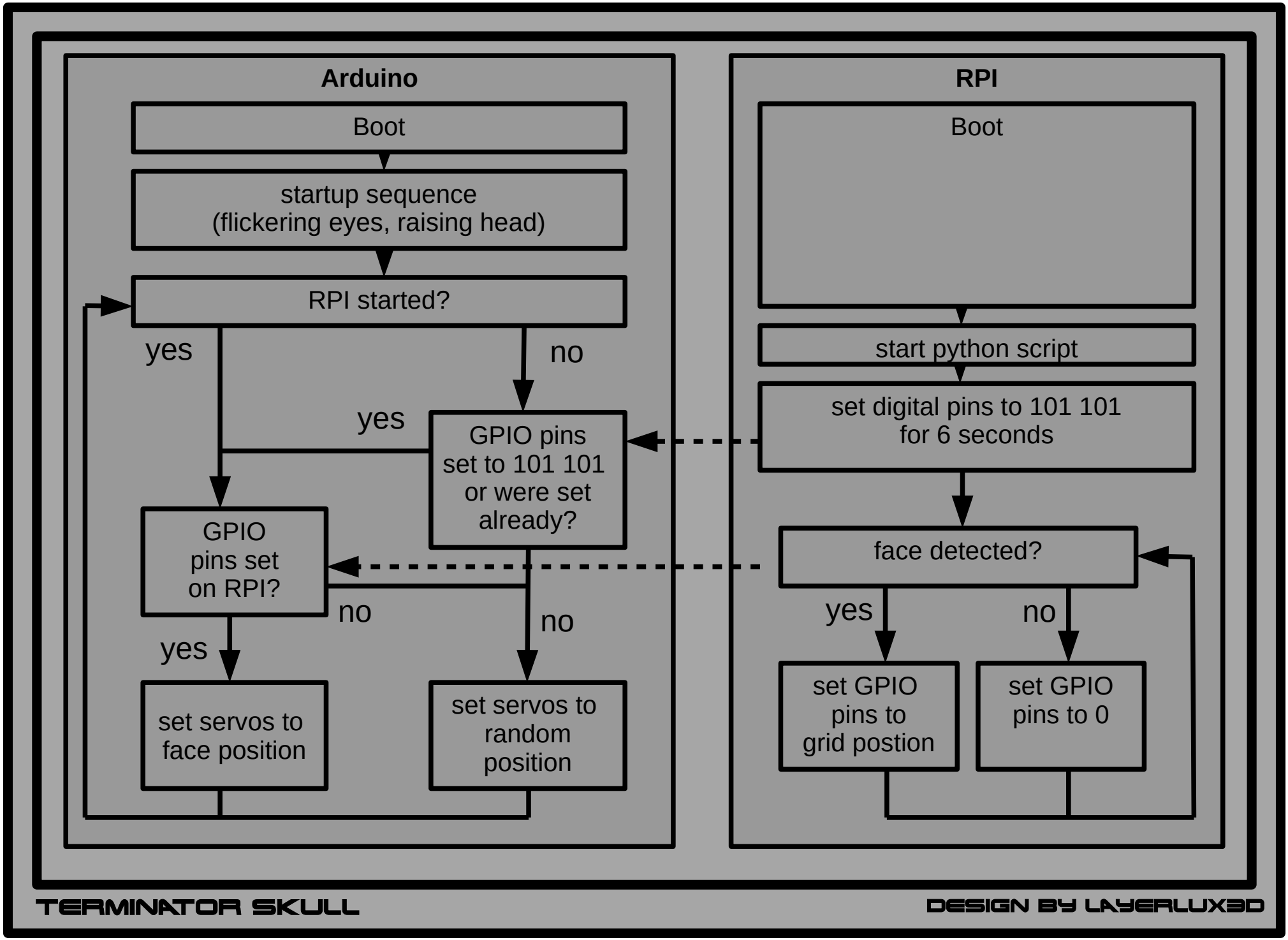
When the RPI has finished booting, it starts the face tracking script and sets its GPIO pins to 101 101 (the ready signal). This configuration signals to the Arduino that the RPI's startup is complete and from now on, the GPIO pins will be set to values that can be interpreted.

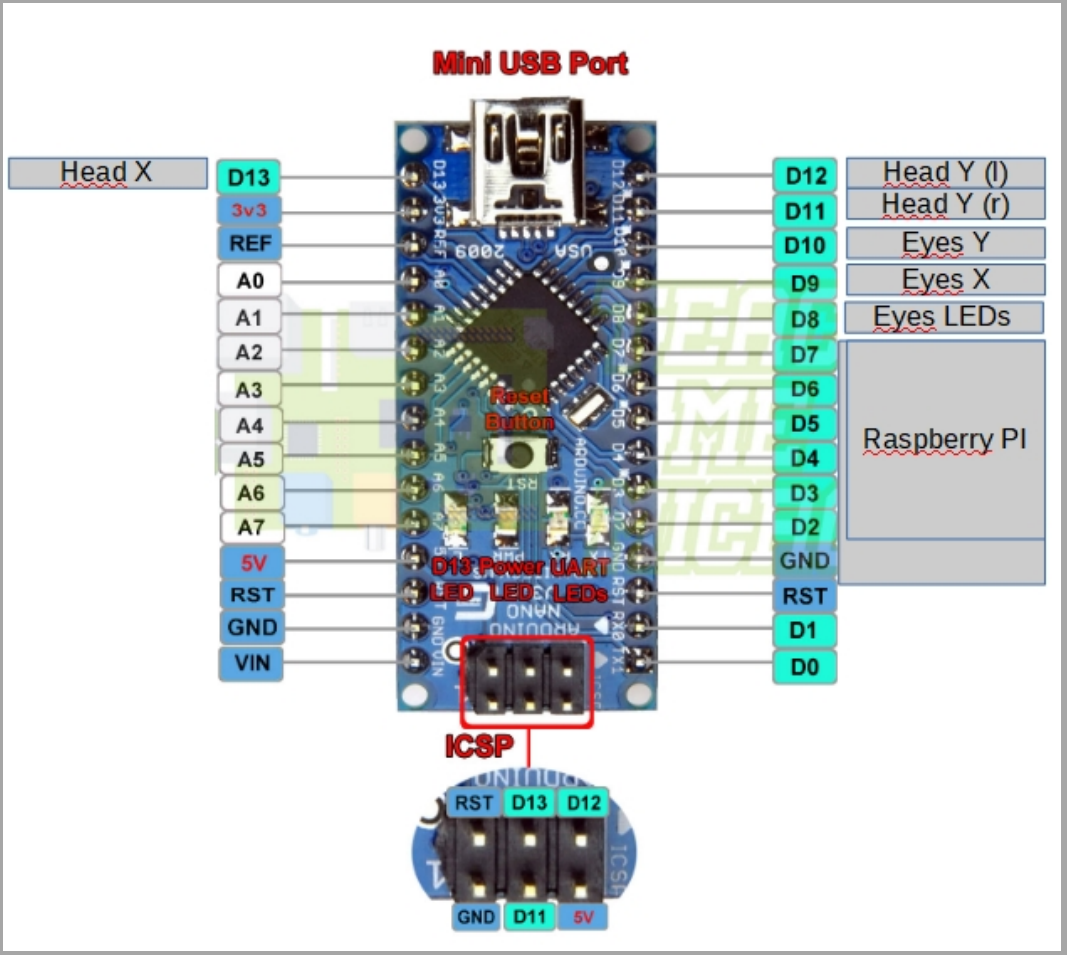
When the Arduino reads the ready signal, the eyes turn off for a moment and then turn on again. This indicates that the RPI has started successfully and will send face tracking data, and the Arduino will respond accordingly.

The RPI checks for a face position. If a face is detected, the screen is divided into a 7x7 grid, and the position in the grid is set via binary code to the 6 GPIO pins (3 pins for x, 3 pins for y).

When the Arduino reads the pins and all pins are set to 0, or if the RPI has not set its ready signal, the Arduino sets the servos to a random position.

If the pins are set to a face position, the Arduino moves the servos in the direction of the face.


Arduino
Boot
startup sequence
(flickering eyes, raising head)
RPI started?
yes
no
yes
GPIO pins
set to 101 101
or were set
already?
GPIO
pins set
on RPI?
no
no
yes
set servos to
face position
set servos to
random
position
RPI
Boot
start python script
set digital pins to 101 101
for 6 seconds
face detected?
yes
no
set GPIO
pins to
grid postion
set GPIO
pins to 0
TERMINATOR SKULL
DESIGN BY LAYERLUX3D

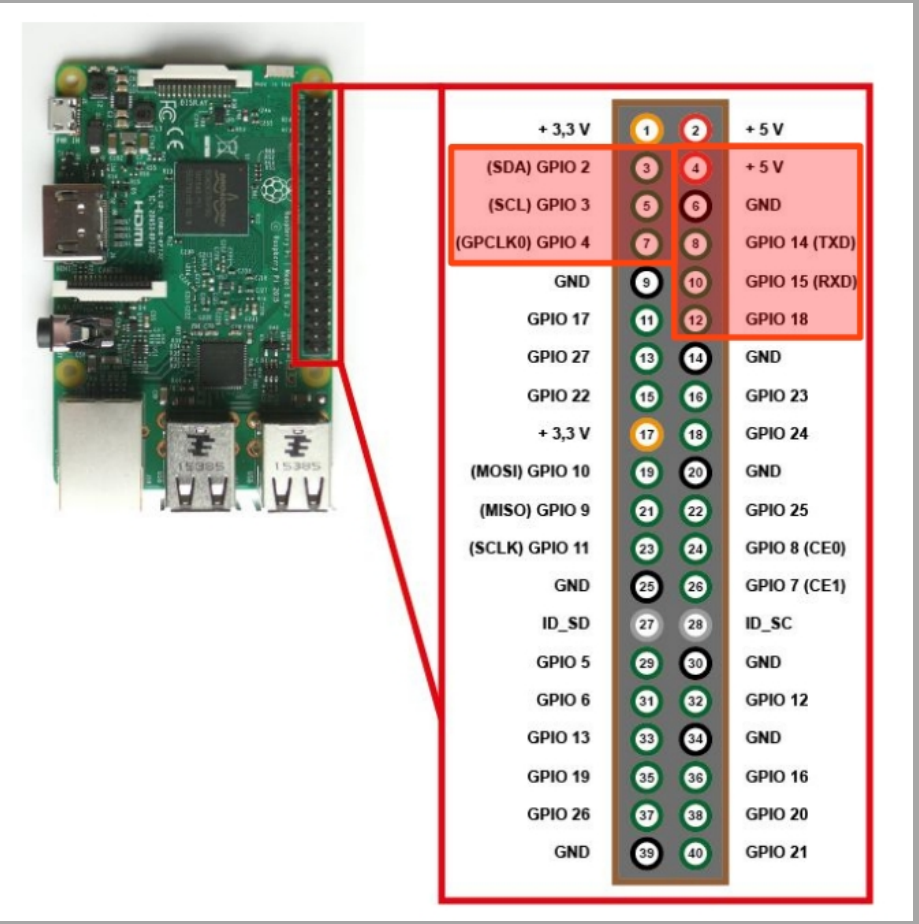


There is a new version of the Arduino code.

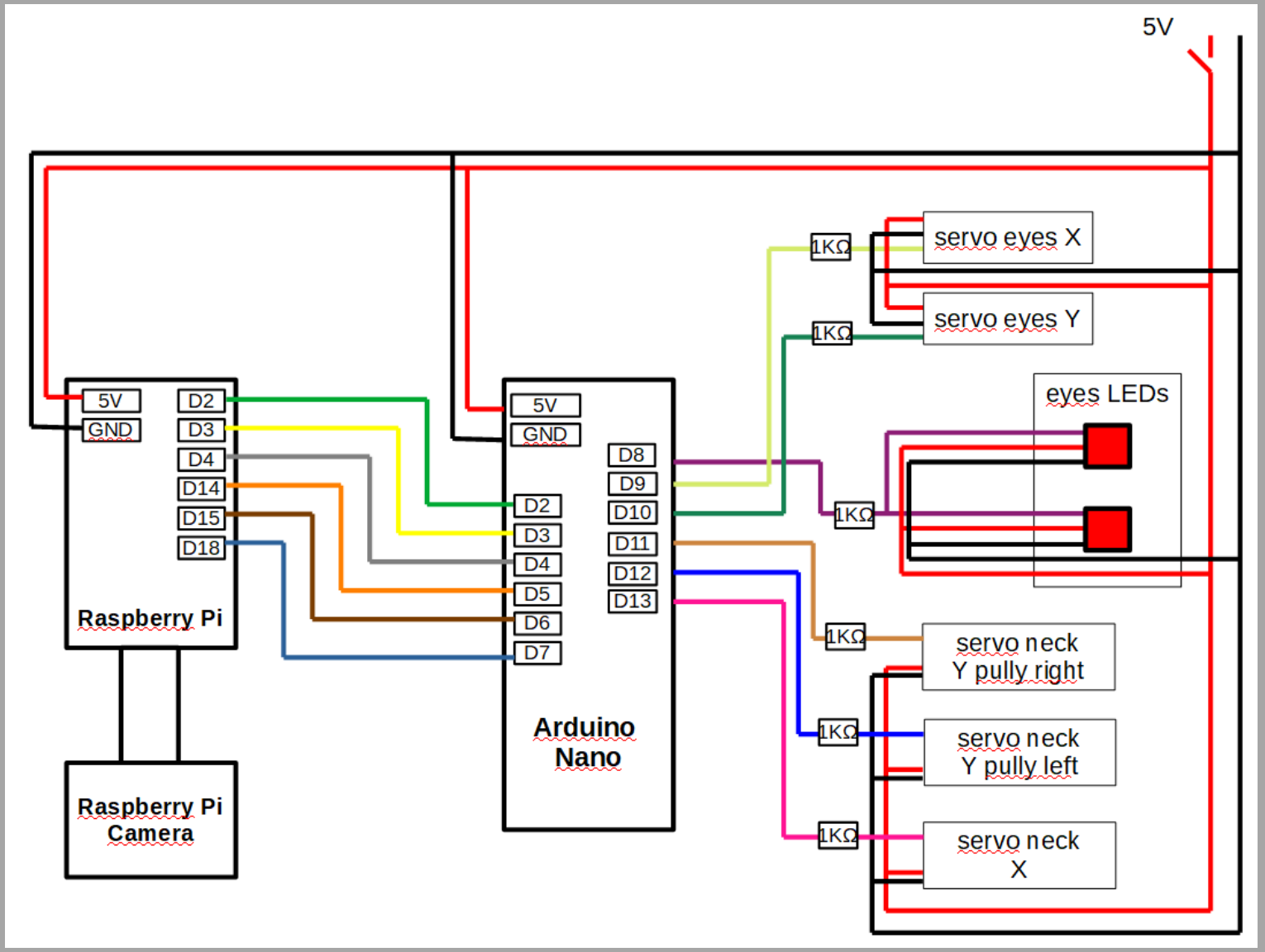
terminator_arduino_facetracking.ino

The pins have chaned compaired to the previous version! You can see the changes in the schematics.

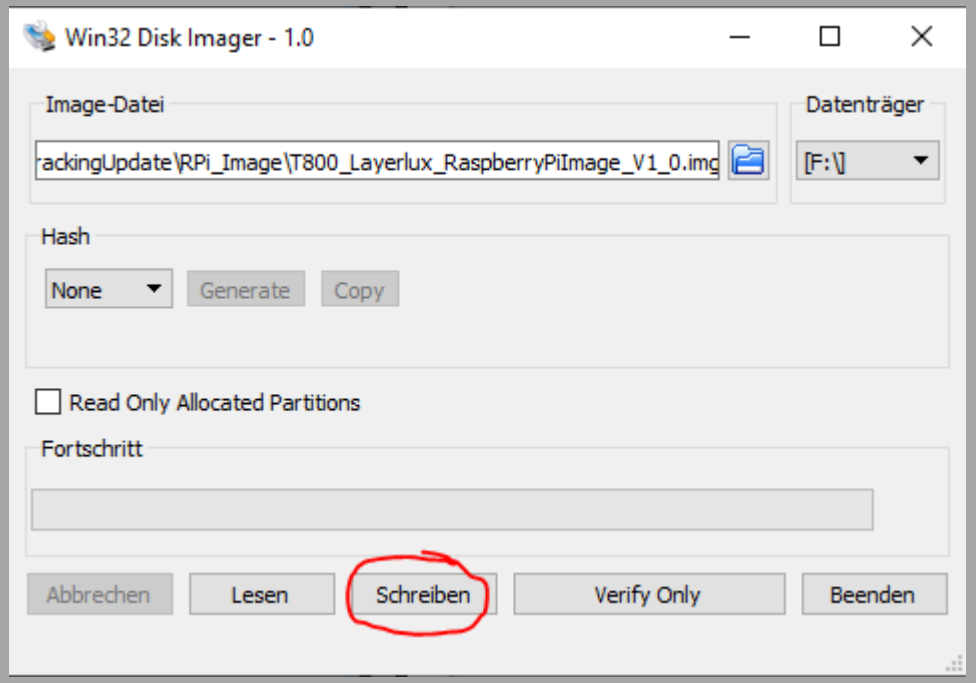
picture source
https://www.teachmemicro.com/arduino-nano-pinout-diagram/



picture source
https://www.elektronik-kompendium.de/sites/raspberry-pi/1907101.htm



Please note! The pin connections between the LEDs/servos to the Arduino have changed compaired to the non RPI version!



Load the image T800_Layerlux_RaspberryPiImage_V1_0.img to the SD card with the programm Win32 Disc Imager.

Note that depending on the SD card the boot process of the Raspberry Pi is faster or slower.

You can connect a HDMI cable to the RPI to check out what is happening on it.

When the RPI starts up it starts the Python programm *facetracker_new.py* that is located on the desktop.

A window pops up and you can see what the cam sees. That way you can adjust the camera if needed.

I noticed that the RPI can get hot while running because the face tracking operation is very demanding on the CPU.

In the python programm you can find two places in the code with ‚# hot‘

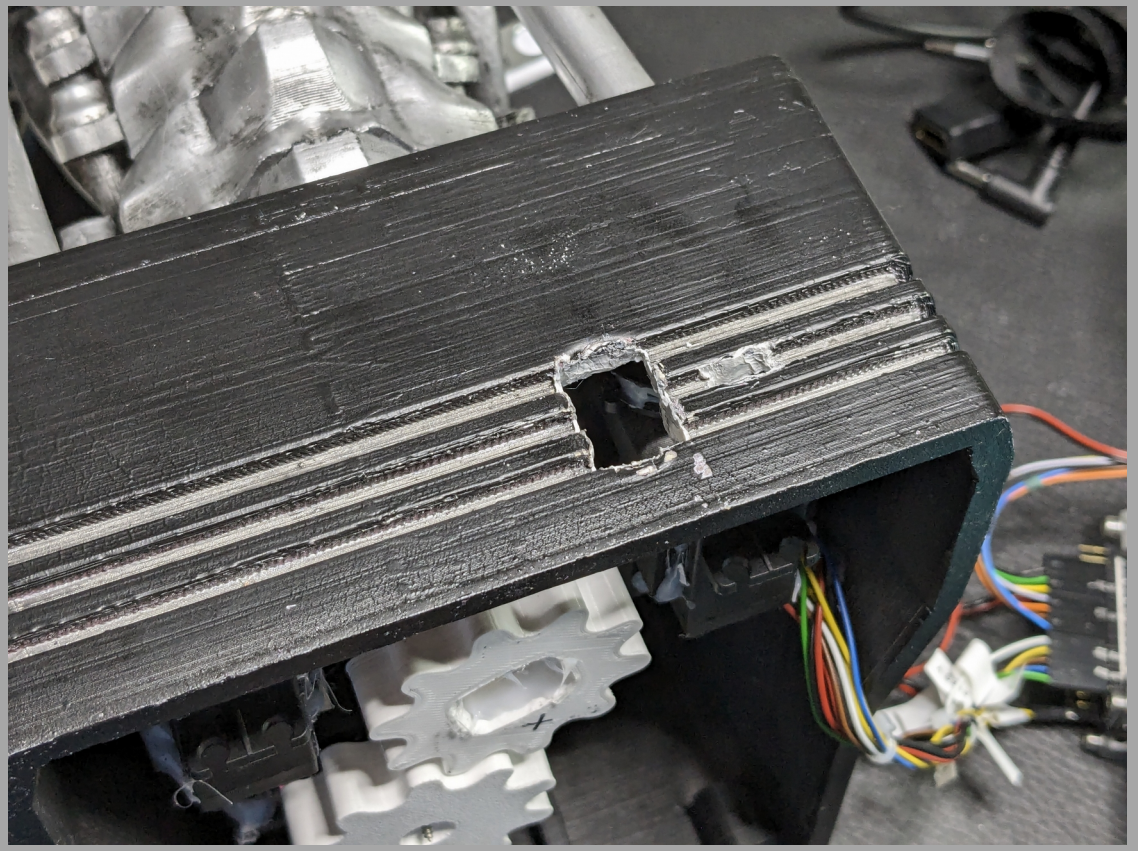
I reduced the resolution of the camera here. When you increse the resolution the face tracking works from a bigger distance.

```
# hot
#size = 500
size = 300
```

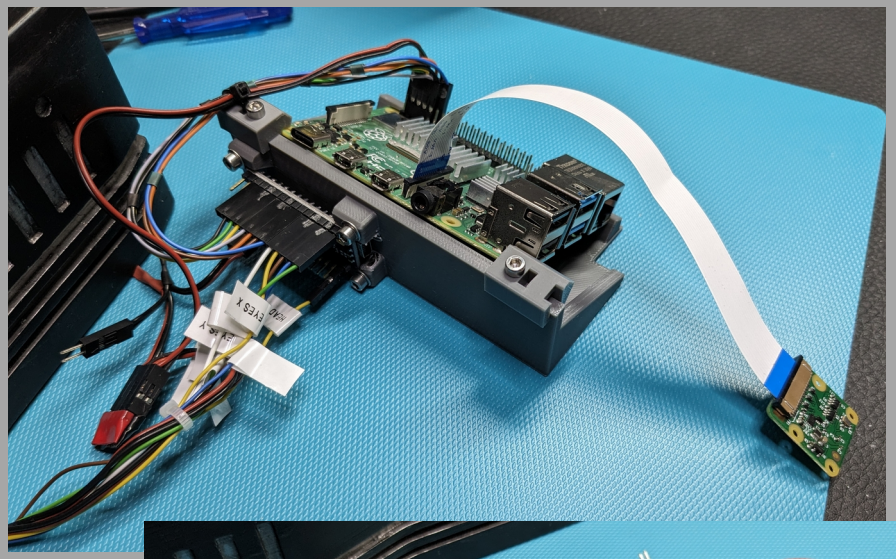
And I build in a small delay so that the CPU is ‚idle‘ for a short amount of time.

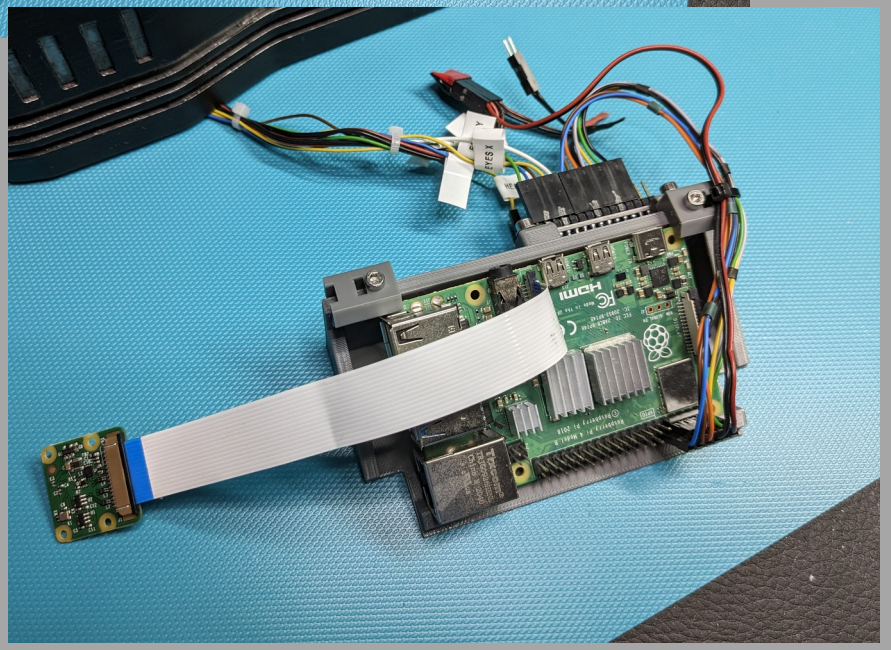
```
# hot
sleep(0.1)
```

If you have active cooling or a different RPI that does not have that issue you can tweak the settings.



You can cut holes for the power supply and switch in the back with a dremel or just heat up a screw driver and burn the holes that way.





Install the Raspberry Pi and Arduino Nano on the new files 39_pi_holder.stl and 40_pi_holder_M.stl.

You can fix the components with M3 screws.

The cables can be fixed with a zip tie and the holes in 40_pi_holder_M.stl.



Install the USB port, switch and holder in the base and fix it with some hot glue for example.

Pay attention that the cables don‘t interfere with the gears.

Congratulations! You upgraded your T-800 animatronic to face tracking!

Please let me know when you updated it successfully or if you had any trouble. I‘ll update the instructions or the files than.

You can conted me via social media or layerlux3d@protonmail.com

Thanks!

LayerLux3D